

Firewall Configuration

Cybersecurity Portfolio Project

1. Project Overview

This project focused on configuring and managing Windows Defender Firewall settings to improve network security and control inbound traffic. The lab demonstrated how firewall rules help protect systems from unauthorized access by filtering network communication based on ports and protocols.

2. Objective

The goal of this project was to configure Windows Defender Firewall settings, enable firewall protection, and create inbound firewall rules to control network traffic for specific services and ports.

3. Tools Used

- Windows Defender Firewall
- Windows Defender Firewall with Advanced Security
- Windows 10
- VirtualBox

4. Skills Demonstrated

- Firewall Configuration
- Network Security
- Port Management
- Access Control
- Windows Administration
- Inbound Rule Creation
- Security Hardening

5. Lab Environment

The lab environment was created using a Windows 10 virtual machine inside VirtualBox. Windows Defender Firewall and Windows Defender Firewall with Advanced Security were used to configure network protection settings and manage inbound firewall rules.

6. Steps Performed

1. Opened Windows Defender Firewall settings.
2. Reviewed firewall protection status for public and private networks.
3. Enabled Windows Defender Firewall protections.
4. Configured firewall notification settings.
5. Opened Windows Defender Firewall with Advanced Security.
6. Created a new inbound firewall rule.
7. Configured TCP port 21 for FTP traffic.
8. Verified that the inbound firewall rule was successfully created and enabled.

7. Screenshots

Figure 1 — Windows Defender Firewall Status

- Reviewed current firewall protection settings.
- Observed firewall disabled status for public networks.

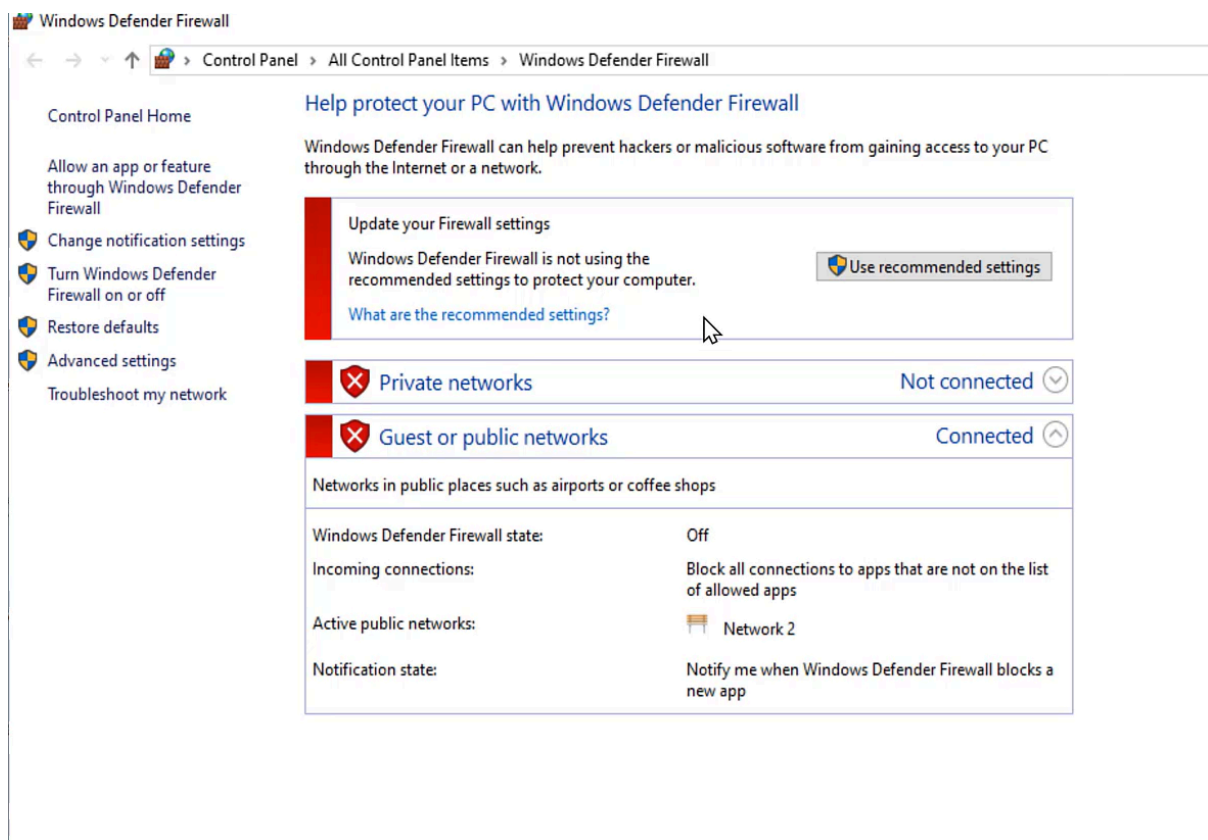


Figure 2 — Enabling Firewall Protection

- Enabled Windows Defender Firewall for:
 - Private networks
 - Public networks
- Configured notification settings.

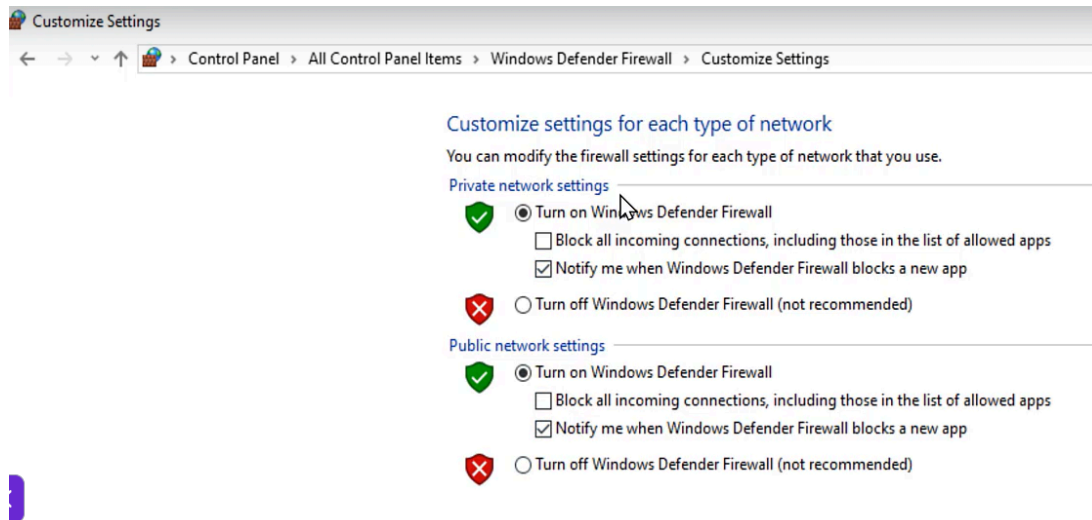


Figure 3 — Creating an Inbound Firewall Rule

- Opened the New Inbound Rule Wizard.
- Configured a TCP rule for port 21 (FTP).

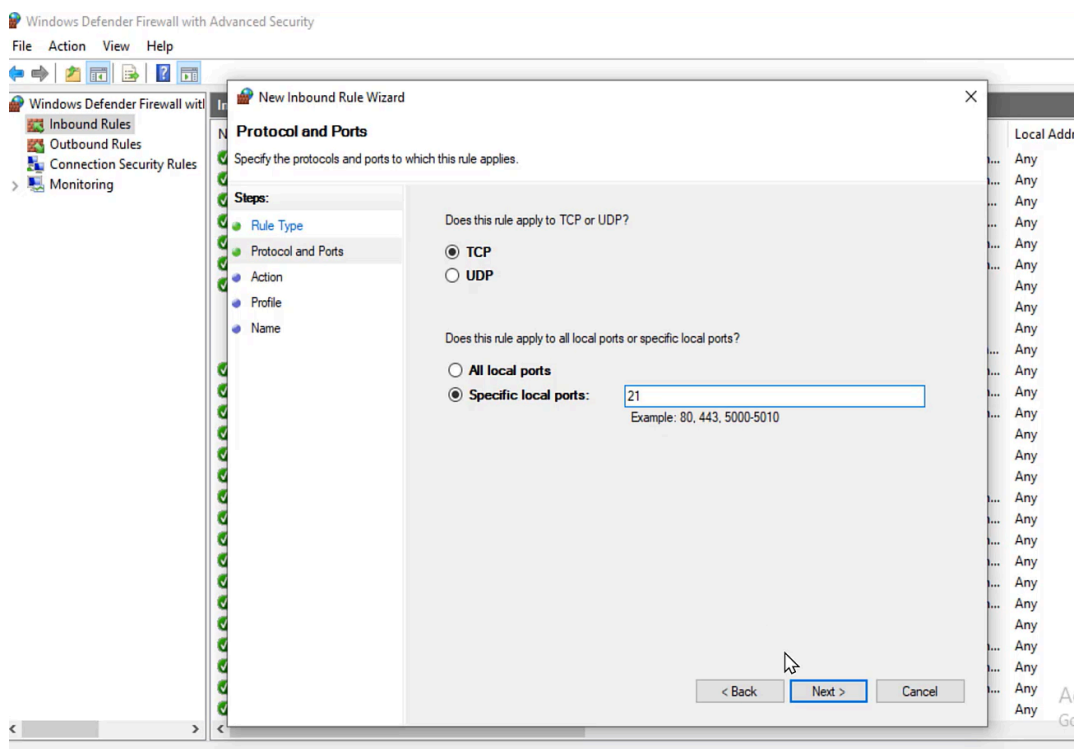
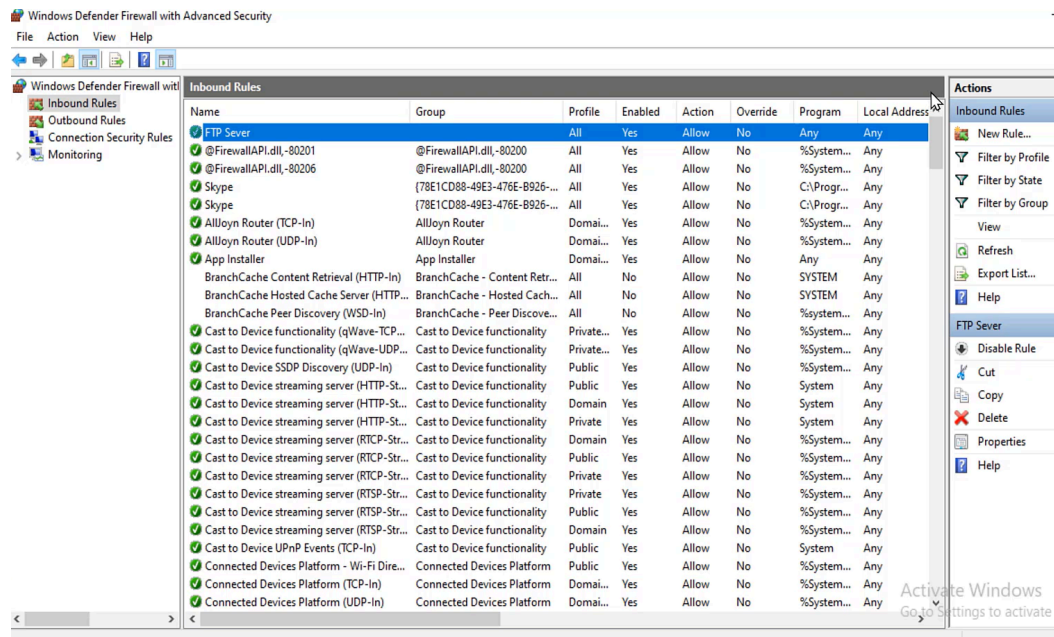


Figure 4 — Verifying Firewall Rule Creation

- Verified that the “FTP Server” inbound rule was successfully enabled.
- Confirmed firewall configuration inside Windows Defender Firewall with Advanced Security.



8. Security Benefits

Firewalls help protect systems by filtering unauthorized network traffic and controlling access to services and ports.

Firewall configuration helps:

- Reduce attack surfaces
- Control inbound traffic
- Block unauthorized access
- Improve network security
- Protect exposed services

9. What I Learned

I learned how Windows Defender Firewall can be configured to manage network access and secure system communication through inbound rules and port filtering.

I also gained experience creating firewall rules for specific services and understanding how firewalls help enforce network security policies.

10. Conclusion

This project demonstrated practical cybersecurity skills related to firewall configuration, network protection, and access control using Windows Defender Firewall and Advanced Security settings.